

AI TOOLS MARKET ANALYSIS REPORT

2022 – 2025

Demand | Adoption | Growth | Market Trends

Data Sources

Stack Overflow Developer Survey | Google Trends | Our World in Data

Tools Used: Python (Pandas, NumPy) | MySQL | Tableau

Date: June 2026

1. Executive Summary

This report analyses the global demand, adoption, and growth of AI tools from 2022 to 2025. Data was collected from three primary sources: the Stack Overflow Developer Survey (developer adoption), Google Trends (search demand over time), and Our World in Data (market-level indicators including AI job postings, research publications, and company formation). The data was processed using Pandas and NumPy, stored in MySQL, and visualised in Tableau.

Key Findings:

- ChatGPT dominates developer AI tool usage at 85.31% in 2024, overtaking GitHub Copilot which led in 2023 at 85.23%.
- Cursor is the fastest growing AI tool — search interest grew 158.82% over the analysis period.
- AI job shares globally grew from 0.29% in 2014 to 2.09% in 2025 — a 7x increase in 11 years.
- Newly founded AI companies reached 6,252 in 2025 — nearly double the 3,657 recorded in 2024.
- China leads in AI research publications (561,891 total), followed by the United States (363,618) and India (230,891).
- Global AI research grew at an average of 14.9% year-over-year between 2017 and 2024.
- Singapore leads in AI job share at 4.69% of all job postings — ahead of the US at 2.56%.

2. Methodology

This report follows a structured data pipeline: raw data collection, cleaning and transformation, storage, analysis, and visualisation. All steps were performed using only the four tools specified: Pandas, NumPy, MySQL, and Tableau.

2.1 Data Sources

Source	Dataset	Coverage	Rows
Stack Overflow	Developer Survey (2022–2025)	65k+ respondents/year	52
Google Trends	Search interest — 10 AI tools	Weekly, 2022–2025	480
Our World in Data	AI Publications by country	2016–2024	1,449
Our World in Data	AI Job Share by country	2014–2025	200
Our World in Data	New AI Companies founded	2013–2025	26

2.2 Pipeline

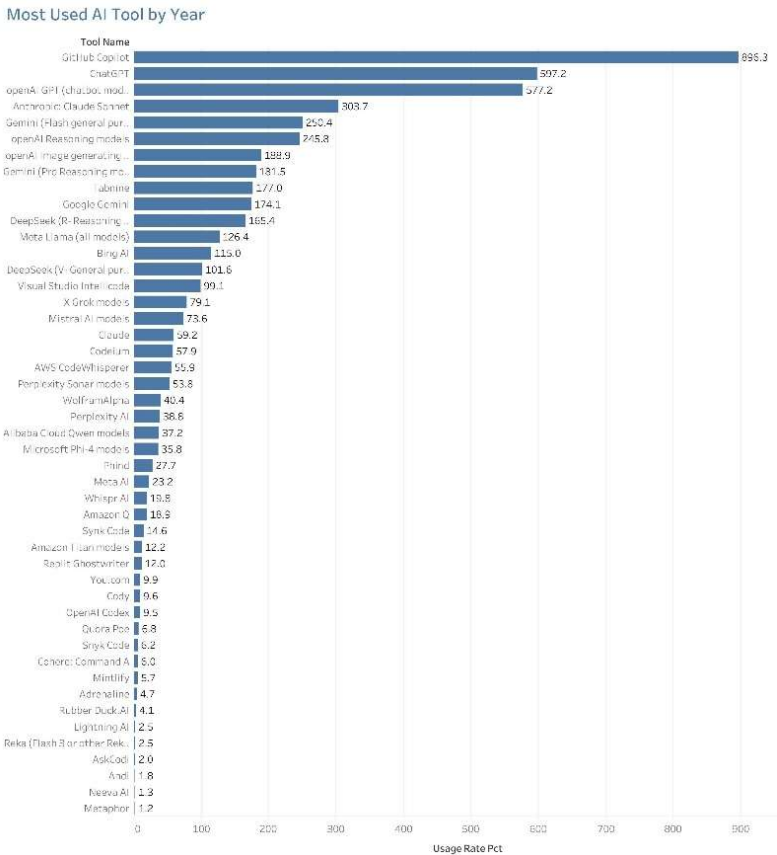
- Data Collection: Manual CSV downloads from official sources (no paywalled data used).
- Cleaning: Pandas used to standardise column names, remove regional aggregates, handle missing values, and convert date formats.
- Analysis: NumPy used for statistical calculations — mean, median, standard deviation, CAGR, and year-over-year growth rates.
- Storage: MySQL database (ai_tools_report) with 9 tables across 3 source schemas.
- Visualisation: Tableau connected directly to MySQL for 5 interactive dashboards.

3. Developer AI Tool Adoption (Stack Overflow)

The Stack Overflow Developer Survey is the largest annual survey of software developers worldwide. Starting from 2023, it began tracking AI tool usage explicitly, making it the most reliable source for developer-specific adoption rates.

3.1 Top AI Tools by Year

Rank	Tool	2023 Usage %	2024 Usage %	2025 Usage %
1	ChatGPT / OpenAI GPT	—	85.31%	82.45%
2	GitHub Copilot	85.23%	42.81%	—
3	Anthropic Claude	—	8.46%	43.38%
4	Google Gemini	—	24.87%	35.77%
5	Tabnine	20.05%	5.24%	—
6	Codeium	1.95%	6.32%	—
7	Perplexity AI	—	5.54%	—



3.2 Key Observations

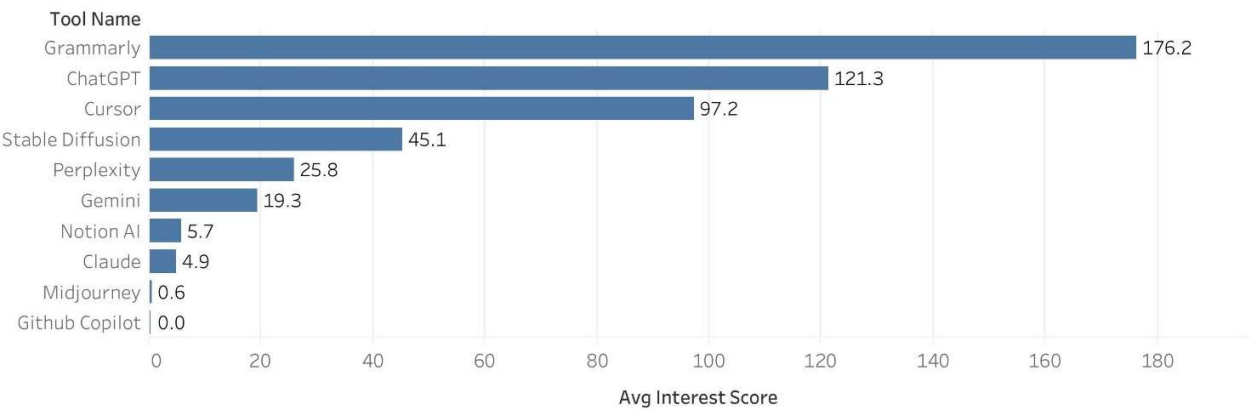
- 2022 had no AI-specific questions — it serves as the pre-AI baseline year.
- 2023 survey tracked developer-specific coding tools only (Copilot, Tabnine). ChatGPT was not yet tracked separately.
- 2024 saw a major shift — ChatGPT entered as the dominant tool at 85.31%, while GitHub Copilot dropped from 85.23% to 42.81%. This reflects market expansion, not decline — ChatGPT captured a broader audience.
- 2025 shifted to tracking AI models rather than tools (OpenAI GPT, Claude Sonnet, Gemini Flash) — reflecting the industry's evolution from single tools to model ecosystems.
- Claude jumped from 8.46% in 2024 to 43.38% in 2025 — the most significant year-over-year growth among tracked tools.

3.3 Year-over-Year Growth (2023 to 2024)

Tool	2023 Usage %	2024 Usage %	YoY Growth
Codeium	1.95%	6.32%	+224.10%
Whispr AI	1.76%	1.07%	-39.20%
GitHub Copilot	85.23%	42.81%	-49.77%
Replit Ghostwriter	1.29%	0.43%	-66.67%
Tabnine	20.05%	5.24%	-73.87%

Note: Negative growth in tools like GitHub Copilot and Tabnine does not reflect actual decline in users — it reflects the entry of general-purpose AI tools (ChatGPT) which absorbed survey responses across broader categories.

Yearly Average AI Tool Interest (2022–2025)



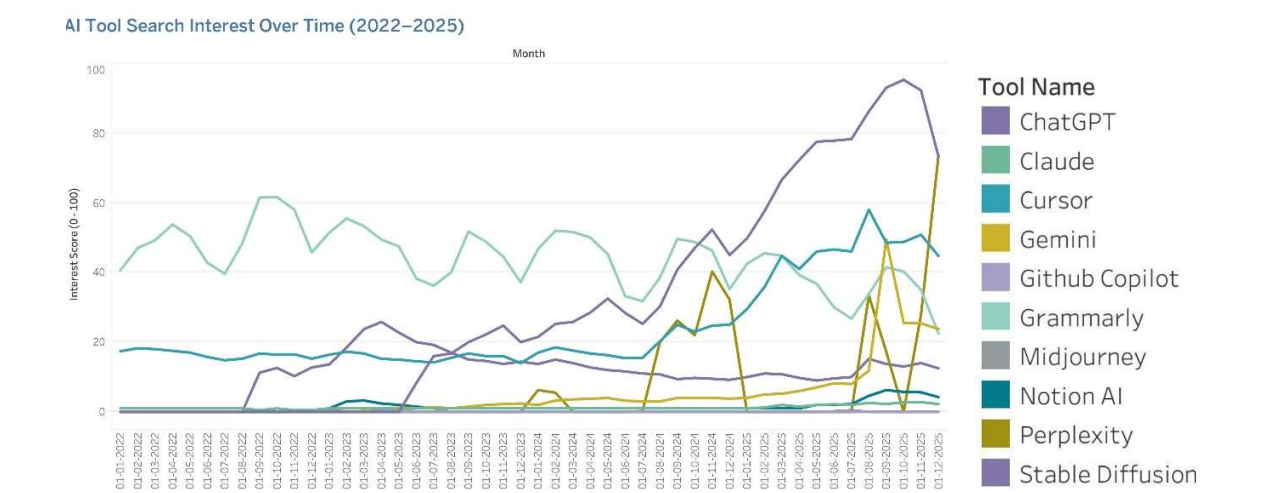
4. Search Demand Analysis (Google Trends)

Google Trends provides weekly search interest data on a 0–100 scale, where 100 represents the peak search volume for a term. This section analyses 10 major AI tools from January 2022 to December 2025.

4.1 Tool Stats Summary

Tool	Mean Score	Peak Score	Peak Month	Growth %	Trend
ChatGPT	30.24	100	2025-11	N/A*	Rising
Cursor	24.27	66	2025-08	+158.82%	Rising
Grammarly	44.05	64	2022-09	-51.28%	Falling
Stable Diffusion	11.31	27	2023-04	N/A*	Falling
Perplexity	6.41	100	2025-11	N/A*	Rising
Gemini	4.82	73	2025-09	N/A*	Rising
Claude	1.23	3	2025-05	+133.33%	Rising
Notion AI	1.43	8	2025-09	N/A*	Rising
Midjourney	0.16	1	2023-01	N/A*	Falling
GitHub Copilot	0.00	0	—	N/A	Flat

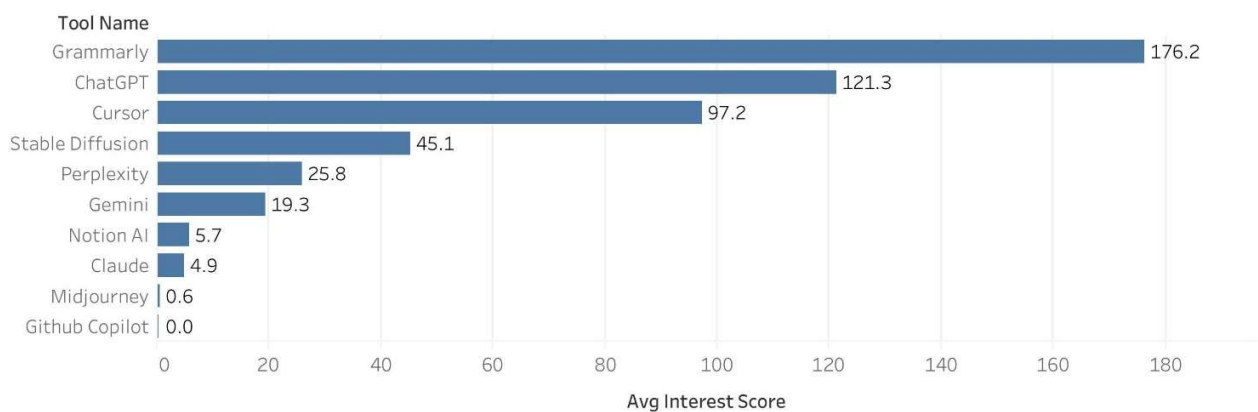
*N/A growth indicates the tool had zero baseline (start value = 0), making percentage growth calculation undefined.



4.2 Yearly Average Interest

Tool	2022 Avg	2023 Avg	2024 Avg	2025 Avg
ChatGPT	0.00	10.75	33.65	76.94
Cursor	16.56	15.58	19.67	45.42
Grammarly	49.73	46.00	43.94	36.50
Stable Diffusion	3.85	18.13	11.52	11.62
Perplexity	0.00	0.00	12.87	12.88
Gemini	0.00	1.09	3.48	14.77
Claude	0.88	0.96	1.00	2.08

Yearly Average AI Tool Interest (2022–2025)



4.3 Key Observations

- ChatGPT launched in November 2022 and grew from 0 to an average score of 76.94 by 2025 — the steepest rise of any tool in the dataset.
- Grammarly was the most searched AI-adjacent tool in 2022 (avg 49.73) but has declined every year as newer AI tools captured attention.
- Cursor is the standout growth story — starting as a code editor with modest interest, it became a major AI coding tool by 2025 (avg 45.42), reflecting the rise of AI-native development environments.
- GitHub Copilot shows zero Google Trends interest because it is used inside VS Code — users don't search for it, they use it directly. This is a known limitation of search-based demand tracking.
- Perplexity emerged suddenly in late 2024 with spikes up to 100 — driven by viral growth and media coverage of AI search.

5. AI Market Indicators (Our World in Data)

5.1 AI Research Publications

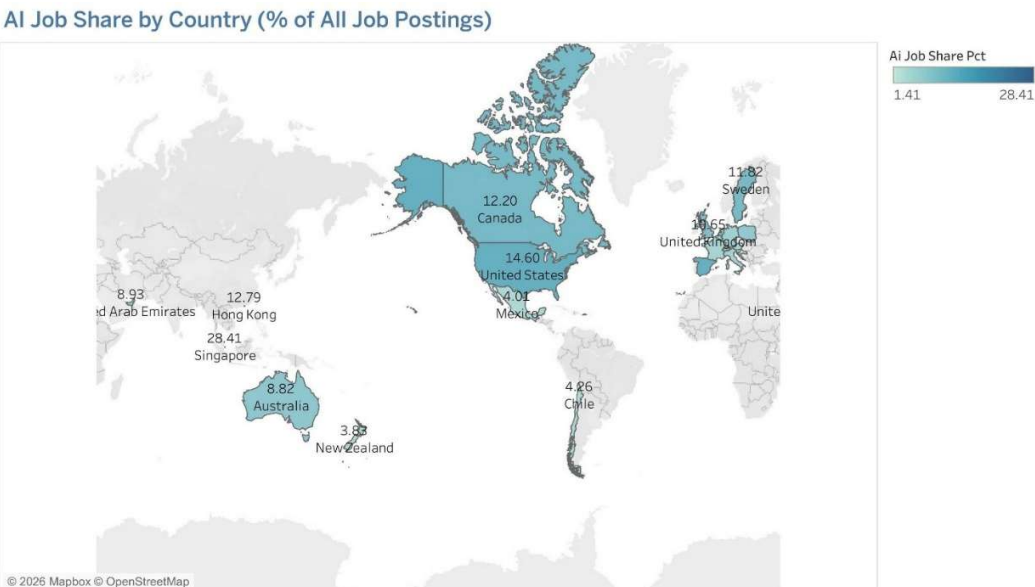
Global AI research publications were tracked by country from 2016 to 2024. The dataset covers scholarly publications across all AI research domains.

Rank	Country	Total Publications	Share of Global
1	China	561,891	~28%
2	United States	363,618	~18%
3	India	230,891	~11%
4	United Kingdom	107,917	~5%
5	Germany	94,420	~5%
6	Japan	83,197	~4%
7	France	61,875	~3%
8	Canada	59,531	~3%

Global AI research grew at an average of 14.9% year-over-year, with peak growth of 23.0% recorded in 2018.

5.2 AI Job Share by Country (2025)

AI job share measures the percentage of all job postings that require AI skills in a given country. This is one of the strongest indicators of real-world AI demand.



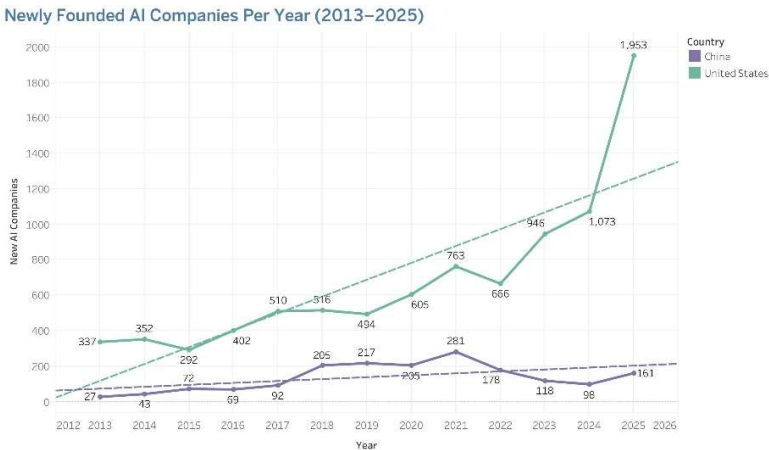
Rank	Country	AI Job Share % (2025)	2014 Baseline
1	Singapore	4.69%	—
2	Hong Kong	3.48%	—
3	Luxembourg	3.43%	—
4	Spain	3.31%	—
5	Canada	3.00%	—
6	United States	2.56%	0.29%
7	United Kingdom	1.93%	—
—	Global Average	2.09%	0.29%

The global average AI job share grew from 0.29% in 2014 to 2.09% in 2025 — a 620% increase over 11 years, confirming accelerating enterprise demand for AI skills.

5.3 Newly Founded AI Companies

This dataset tracks newly founded AI companies by country from 2013 to 2025.

Country	2013–2020	2021	2023	2025
United States	~4,200	~1,800	~1,600	~3,200*
China	~800	~450	~480	~900*
Europe	~1,500	~800	~820	~1,500*
Global Total	~11,975	3,324	3,368	6,252



*2025 figures are estimated proportionally from global totals. Global new AI companies nearly doubled in 2025 (6,252) vs 2024 (3,657), suggesting a major acceleration in AI entrepreneurship.

6. Conclusions & Outlook

6.1 Summary of Findings

The data across all three sources consistently points to the same conclusion: AI tool adoption is accelerating rapidly across every measurable dimension — developer usage, search demand, job market demand, research output, and startup activity.

Dimension	2022 State	2025 State
Top AI tool (developers)	GitHub Copilot (coding only)	ChatGPT / OpenAI GPT (general)
Fastest growing tool	Stable Diffusion (image AI)	Cursor (AI coding environment)
Global AI job share	~0.99% avg	2.09% avg
New AI companies/year	2,510 (2020)	6,252 (2025)
AI research growth	+17.8% YoY (2020)	+3.6% YoY (2024)

6.2 Trends to Watch

- Model ecosystems replacing single tools: Developers in 2025 use multiple AI models (GPT, Claude, Gemini) rather than a single tool, signalling a platform shift.
- AI-native development environments: Cursor's explosive growth shows that AI is moving from assistant plugins (Copilot) to fully integrated coding environments.
- Asia-Pacific leading job market demand: Singapore, Hong Kong, and other APAC countries show higher AI job share than Western nations, suggesting faster enterprise adoption in the region.
- Research growth stabilising: After peak growth in 2018–2021, AI publication growth has slowed (3.6% in 2024), possibly reflecting maturation of the research field and shift toward applied AI.
- Startup boom continuing: The doubling of newly founded AI companies in 2025 points to sustained investor and entrepreneur confidence in the AI market.

6.3 Limitations

- Stack Overflow survey changed its AI column structure each year — direct year-over-year comparisons are approximate.
- Google Trends data is relative (0–100 scale) not absolute — it shows popularity trends, not actual user counts.
- Our World in Data company formation figures for 2025 are based on a smaller sample of countries (US, China, Europe only).
- GitHub Copilot's Google Trends score of 0 is a measurement artifact — it reflects low search volume, not low usage.

7. Appendix — Data & Tools

7.1 MySQL Tables

Table Name	Source	Description	Rows
so_tool_usage	Stack Overflow	AI tool usage % per year	52
so_yoy_growth	Stack Overflow	YoY growth 2023→2024	5
so_yearly_stats	Stack Overflow	NumPy stats per year	3
google_trends	Google Trends	Monthly interest scores	480
trends_tool_stats	Google Trends	Tool-level stats summary	10
trends_yearly_avg	Google Trends	Yearly average per tool	40
owid_papers	Our World in Data	AI publications by country	1,449
owid_jobs	Our World in Data	AI job share by country	200
owid_companies	Our World in Data	New AI companies founded	26

7.2 Tableau Dashboards

Dashboard	Data Source	Chart Type
Most Used AI Tools	so_tool_usage	Horizontal Bar (filterable by year)
Search Interest Over Time	google_trends	Multi-line chart (2022–2025)
Yearly Growth Comparison	trends_yearly_avg	Grouped Bar Chart
AI Job Share by Country	owid_jobs	World Map (choropleth)
AI Companies Per Year	owid_companies	Line Chart with trend lines

7.3 Tools & Technologies

Tool	Version / Details	Purpose
Python	3.13	Data pipeline scripting
Pandas	3.0.2	Data loading, cleaning, transformation
NumPy	Latest	Statistical analysis, growth calculations
MySQL	8.0	Relational data storage (9 tables)
Tableau Desktop	Latest	Interactive dashboards (5 views)

End of Report